# Data analysis of factors related to the fishing performance of Tasmanian skippers

## Aim

The analysis below provides an insight into the factors related to the fishing performances of Tasmanian skippers, such as the weight of caught fish known as 'Catch', and the dollar per hour value of caught fish, colloquially referred to as 'Effectiveness'.

## Background

This observational study examined data about 152 Tasmanian fishing boats. Besides 'Catch' figures, other data collected were the dollar value of caught fish per vessel known as 'Value', skippers' level of fishing experience labelled as 'Experienced', the boat port of origin known as 'Port', the quality of sonar equipment labelled as 'Search', length of fishing trip categorised as 'Time', and lastly, fishermen preference between shallow or deep-water fishing classified as 'Depth'.

## Findings

1. Overall, experienced skippers caught an additional 5,990 to 9,180 kg of fish compared to inexperienced fishermen (Table 1). Similarly, experienced fishers showed an advantage in the value of caught fish per hour. Specifically, experienced anglers were catching fish at a rate of 40.11 to 189.07 $/h compared to inexperienced skippers (Table 1).
2. Furthermore, experienced skippers who used sophisticated search equipment caught on average 2,860 to 7,020 kg more fish than inexperienced fishers who used the same search gear (Table 2). Similarly, experienced anglers with adequate sonars caught 7,190 to 11,370 kg more fish than novice fishermen with adequate sonars (Table 2). These two observations point out that the experienced skippers had caught markedly more fish than inexperienced fishermen despite the type of sonar equipment.
3. The current analysis showed that about 52% to 71% of experienced anglers caught over 42 tonnes of fish. In contrast, only 4.5% to 25.3% of inexperienced fishermen seized over 42 tonnes of fish (Table 3).
4. Boats that employ sophisticated search equipment caught an extra 1,420 to 5,180 kg of fish than fleets that utilised adequate search gear (Table 4).
5. Furthermore, fleets fishing in shallow waters caught on average additionally 850 to 4,665 kilograms of fish than anglers that prefer to fish in deep waters (Table 5).
6. This said, the study looked into the relationship between the use of search equipment among Tasmanian fishers and the depths at which all fleets fish. In sum, an association between all sonar equipment and fishing depths was found which might be worth further investigation (Table 6).

## Statistical methods and results

### Variables

The variable 'Effectiveness' was introduced to operationalise fishermen's productivity. The formula for 'Effectiveness' was =1000*('value'/'time'), where the 'Value' referred to the dollars paid per catch per boat and 'time' referred to the hours each vessel spent at sea.

The variable 'Bincatch' categorised boats as 'high' value if the fishermen caught more than the median value of 42 tonnes. Conversely, the boats were 'low' if they caught less than 42 tonnes. The Excel formula for 'Bincatch' was =IF(B2>N$2, "high", "low"), where N$2 was equivalent to the median 'catch' variable.

### Computational methods

- All statistics and tables were calculated and created in Microsoft Excel 2022.
- T-tests were computed using the 'Two-Sample Assuming Unequal Variances' function as part of the 'Data Analysis' tool pack located under the Excel 'Data' tab.
- The newly developed variable 'Effectiveness' was calculated with Excel following the formula =1000*('Value' column/'Time' column).
- T-test 95% confidence intervals were estimated with the formula =mean differences ± 2*(SQRT*(S2/*n1*+ S2/*n2*)), where S2 was equivalent to the mean variance while *n* was the sample size.
- A 'Pivot Table' was utilised to create a frequency table that compared two categorical variables, 'Experience' and the newly introduced 'BinCatch' variable.
- Proportions for experienced skippers that caught more than 42 tonnes of fish as well as proportions for inexperienced skippers that surpassed the median 'Catch' were calculated in Excel =(counts of over 42 tonnes within experienced or inexperienced category / the total number above and bellow 42 tonnes within the experienced or inexperienced category).
- 95% Confidence Intervals were calculated in Excel following the formula = 'proportion of interest' ± 2*(SQRT*((1-)/*n*)).
- The Chi-Square Test of Independence was calculated in Excel, following the formula $X^2$=Σ [(Observed-Expected count)2/Expected count]. Lastly, the $X^2$ probability was determined following the Excel command =1-CHISQ.DIST(x, deg_freedom, cumulative).

## Results

*Table 1.* Difference in fish tonnage and 'Effectiveness' in relation to skippers' experience

| Variables | | *n* | Mean | Variance | *df* | *t* | *p* | *CI* |
|---|---|---|---|---|---|---|---|---|
| **Catch (tonnes)** | **Inexperienced** | 47 | 37.1 | 19.7 | 95 | 9.47* | *P* <.001 | 5.99, 9.18 |
| | **Experienced** | 105 | 44.6 | 23.0 | | | | |
| **Effectiveness ($/h)** | **Inexperienced** | 47 | 579.5 | 43592.1 | 93 | 3.08** | *P* <.003 | 40.11, 189.07 |
| | **Experienced** | 105 | 694.1 | 48219.2 | | | | |

**A two-tailed t test assuming unequal variances for the difference in the weight of caught fish between inexperienced and experienced skippers. There is strong evidence to reject the null hypothesis, suggesting no difference between the mean value of caught fish and skippers' experience, t(95)= 9.47, p< .001. We are 95% confident that the true population of experienced fishermen will catch between 5.99 and 9.18 tonnes of fish. The 95% Confidence Interval (95% CI) was calculated with a point estimate difference = 7.58 and standard error = 0.80.*
***A result of a two-tailed t-test assuming unequal variances for the differences in 'Effectiveness' between inexperienced and experienced fishermen, t(93)= 3.08, p= 0.003. According to the p-value, there is sufficient evidence to reject the proposition that there is no difference in effectiveness between inexperienced and experienced fishermen. The 95% CI was calculated with a point estimate difference of 114.59 and a standard error of 37.24.*

*Table 2.* Analysis of 'Catch' according to fishermen's experience and quality of sonar equipment

| Variables | | *n* | Mean | Variance | *df* | *t* | *p* | *95% CI* |
|---|---|---|---|---|---|---|---|---|
| **Catch (tonnes) w/ Sophisticated search equipment** | **Inexperienced** | 20 | 40.10 | 14.83 | 37 | 4.77* | p < .001 | 2.86, 7.02 |
| | **Experienced** | 63 | 45.03 | 20.58 | | | | |
| **Catch (tonnes) w/ Adequate search equipment** | **Inexperienced** | 27 | 34.89 | 12.18 | 67 | 8.89 | p < .001 | 7.19, 11.37** |
| | **Experienced** | 42 | 44.17 | 26.83 | | | | |

**The relationship between the number of fish caught by inexperienced and experienced fishermen who used sophisticated or adequate search equipment was tested with a two-tailed t-test assuming unequal variances. There is robust evidence to reject the null premise, which suggests no difference between the weight of fish caught by experienced and inexperienced fishermen who use sophisticated sonars, t(37)= 4.77, p< .001. A 95% CI was computed with a mean difference of 4.93 and a standard error of 1.03.*
***Similarly, we are 95% confident that the true population of experienced fishers will catch caught an extra 7.19 to 11.37 fish than inexperienced skippers who used the same equipment. A 95% CI was calculated with a mean difference of 9.28 and a standard error of 1.04.*

*Table 3.* Difference in catches over 42 tonnes between inexperienced and experienced skippers

| **Level of fishing experience** | ***n*** | **Catch > 42 Tonnes** | **Catch < 42 Tonnes** | **Point estimate= Catch > 42 Tonnes / n*** | **Standard error** | ***95% CI*** |
|---|---|---|---|---|---|---|
| **Inexperienced** | 47 | 7 | 40 | 0.15 | 0.052 | 0.045, 0.253** |
| **Experienced** | 105 | 65 | 40 | 0.62 | 0.047 | 0.52, 0.71*** |

*There was a 0.47 difference in *proportions of experienced and inexperienced fishermen who caught more than 42 tonnes of fish. **We are 95% confident that the true population of experienced fishmongers with a catch over 42 tonnes is 0.52 to 0.71. ***In contrast, we are equally confident that 0.045 to 0.253 of the actual population of inexperienced fishmongers will return to their port with a catch over 42 tonnes.*

*Table 4.* The relationship between caught fish and quality of search equipment

| **Variables** | | ***n*** | **Mean** | **Variance** | ***df*** | ***t*** | ***p*** | ***95% CI*** |
|---|---|---|---|---|---|---|---|---|
| **Catch (Tonnes)** | **Adequate equipment** | 69 | 40.54 | 41.63 | 124 | 3.51 | *P* < .0006* | 1.42, 5.18** |
| | **Sophisticated equipment** | 83 | 43.84 | 23.50 | | | | |

*A two-tailed t-test assuming unequal variances was implemented to analyse the relationship between the weight of caught fish and the quality of search equipment.*
**In sum, there is strong evidence to reject the notion that there is no difference in 'Catch' between a fleet that utilises adequate or sophisticated search equipment. We are 95% confident that the actual population of boats that use sophisticated sonar equipment will yield an extra 1.42 to 5.18 tonnes of fish than boats that use adequate equipment, t(124)= 3.52, p< .0006.*
***The CI was calculated utilising a point estimate difference of 3.3 and a standard error of 0.94.*

*Table 5.* The difference in 'Catch' between deep or shallow water fishing

| Variables | | *n* | Mean | Variance | *df* | *t* | *p* | *95% CI* |
|---|---|---|---|---|---|---|---|---|
| **Catch (Tonnes)** | **Deep** | 51 | 40.51 | 29.97 | 106 | 2.87 | P < .005* | 0.85, 4.665** |
| | **Shallow** | 101 | 43.27 | 34.14 | | | | |

*A two-tailed t-test assuming unequal variances was utilised to analyse the relationship between the amount of caught fish and fishing depth. *In general, there is moderate evidence to reject the null, which suggests no association between the water depth and the amount of caught fish, t*(106)= 2.87, *p< .005. **Moreover, we are 95% confident that the true population of boats that practise shallow water fishing will catch between 0.85 and 4.665 tonnes of fish. The 95% CI was calculated utilising a mean difference of 2.76 and a standard error of 0.96.*

*Table 6. Analysis of the association between the quality of search equipment and fishing depths*

| | Variables | Fishing depth | | | | *N* | $X^2$ | *Df* | *p* |
|---|---|---|---|---|---|---|---|---|---|
| | | Deep | | Shallow | | | | | |
| | | ($f_k$) | ($f_e$) | ($f_k$) | ($f_e$)** | | | | |
| **Search equipment** | **Adequate equipment** | 30 | 23.15 | 39 | 45.84 | 152 | 5.58 | 1 | 0.018 |
| | **Sophisticated equipment** | 21 | 27.85 | 62 | 55.15 | | | | |

**A Chi-Square Test of Independence was performed to assess the relationship between the general use of search equipment and the range of fishing depths. The statistics suggest strong to moderate evidence to reject the null hypothesis that implies a similarity in values between the expected of observed counts for search equipment and 'Depth', $X^2$(1, N= 152)= 5.58, p= 0.018. In sum, test results indicate that search equipment and 'Depth' are associated variables.*
***($f_k$) observed counts, ($f_e$) expected counts.*

*Index table A:* Chi-Square test results

| Variables | *N* | $X^2$ | *df* | *p* – one way |
|---|---|---|---|---|
| **Search equipment vs. Experience** | 152 | 3.99 | 1 | 0.046 |
| **Search equipment vs. Depth** | 152 | 5.58 | 1 | 0.018 |
| **Search equipment vs. Port** | 152 | 0.22 | 1 | 0.640 |

*Index table B:* All t-test calculations

| | | *n* | Mean | Variance | *df* | *t* | *s.e.* | *p* – two tail | *95% CI* | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Catch vs. experience** | **Inexperienced** | 47 | 37.11 | 19.79 | 95 | 9.47 | 0.80 | p < .001 | 5.98 | 9.18 |
| | **Experienced** | 105 | 44.69 | 23.03 | | | | | | |
| **Catch vs. depth** | **Deep** | 51 | 40.51 | 29.97 | 106 | 2.87 | 0.96 | 0.005 | 0.83 | 4.68 |
| | **Shallow** | 101 | 43.27 | 34.14 | | | | | | |
| **Catch vs. port** | **East** | 130 | 42.04 | 33.90 | 29 | 1.56 | 1.34 | 0.130 | -4.79 | 0.59 |
| | **West** | 22 | 44.14 | 34.03 | | | | | | |
| **Catch vs. Search equipment** | **Adequate** | 69 | 40.54 | 41.63 | 124 | 3.51 | 0.94 | 0.001 | 1.42 | 5.19 |
| | **Sophisticated** | 83 | 43.84 | 23.50 | | | | | | |
| **Effectiveness vs. Experience** | **Inexperienced** | 47 | 579.59 | 43592.14 | 93 | 3.08 | 37.24 | 0.003 | 40.11 | 189.07 |
| | **Experienced** | 105 | 694.18 | 48219.20 | | | | | | |
| **Effectiveness vs. Depth** | **Deep** | 51 | 644.61 | 59471.51 | 89 | 0.53 | 40.09 | 0.600 | -101.47 | 58.91 |
| | **Shallow** | 101 | 665.89 | 44574.51 | | | | | | |
| **Effectiveness vs. Port** | **East** | 130 | 660.10 | 50308.41 | 29 | 0.19 | 49.54 | 0.850 | -89.76 | 108.39 |
| | **West** | 22 | 650.79 | 45474.11 | | | | | | |
| **Effectiveness vs. Search equipment** | **Adequate** | 69 | 479.86 | 15769.14 | 148 | 13.71 | 23.89 | p < .001 | 279.82 | 375.39 |
| | **Sophisticated** | 83 | 807.47 | 28418.00 | | | | | | |
| **Catch ( tonnes) w/ Sophisticated Search** | **Inexperienced** | 20 | 40.10 | 14.83 | 37 | 4.77 | 1.03 | p < .001 | 2.86 | 7.00 |
| | **Experienced** | 63 | 45.03 | 20.58 | | | | | | |
| **Catch (tonnes) w/ Adequate Search** | **Inexperienced** | 27 | 34.89 | 12.18 | 67 | 8.89 | 1.04 | p < .001 | 7.19 | 11.37 |
| | **Experienced** | 42 | 44.17 | 26.83 | | | | | | |
| **Catch ( tonnes) w/ Sophisticated Search** | **Deep** | 21 | 42.14 | 23.63 | 34 | 1.87 | 1.22 | 0.070 | -4.72 | 0.16 |
| | **Shallow** | 62 | 44.42 | 22.51 | | | | | | |
| **Catch (tonnes) w/ Adequate Search** | **Deep** | 30 | 39.37 | 32.10 | 67 | 1.36 | 1.52 | 0.180 | -5.10 | 0.97 |
| | **Shallow** | 39 | 41.44 | 48.09 | | | | | | |
| **Catch ( tonnes) w/ Sophisticated Search** | **East** | 72 | 43.71 | 22.13 | 12 | 0.55 | 1.86 | 0.590 | -4.74 | 2.70 |
| | **West** | 11 | 44.73 | 34.62 | | | | | | |
| **Catch (tonnes) w/ Adequate Search** | **East** | 58 | 39.97 | 41.26 | 15 | 1.79 | 2.00 | 0.090 | -7.58 | 0.42 |
| | **West** | 11 | 43.55 | 36.07 | | | | | | |
| **Effectiveness ($/h) w/ Sophisticated Search** | **Inexperienced** | 20 | 755.44 | 33635.66 | 29 | 1.50 | 45.79 | 0.150 | -160.12 | 23.03 |
| | **Experienced** | 63 | 823.98 | 26126.86 | | | | | | |
| **Effectiveness ($/h) w/ Adequate Search** | **Inexperienced** | 27 | 449.34 | 11140.18 | 64 | 1.73 | 29.04 | 0.090 | -108.22 | 7.93 |
| | **Experienced** | 42 | 499.48 | 18081.22 | | | | | | |
| **Effectiveness ($/h) w/ Sophisticated Search** | **Deep** | 21 | 869.81 | 34616.05 | 30 | 1.84 | 45.31 | 0.080 | -7.16 | 174.06 |
| | **Shallow** | 62 | 786.35 | 25060.75 | | | | | | |
| **Effectiveness ($/h) w/ Adequate Search** | **Deep** | 30 | 486.97 | 16235.01 | 62 | 0.41 | 30.74 | 0.680 | -48.90 | 74.08 |
| | **Shallow** | 39 | 474.39 | 15757.89 | | | | | | |
| **Effectiveness ($/h) w/ Sophisticated Search** | **East** | 72 | 807.16 | 29944.03 | 15 | 0.05 | 47.67 | 0.960 | -97.62 | 93.05 |
| | **West** | 11 | 809.45 | 20419.98 | | | | | | |
| **Effectiveness ($/h) w/ Adequate Search** | **East** | 58 | 477.53 | 15322.31 | 13 | 0.32 | 45.33 | 0.750 | -105.25 | 76.06 |
| | **West** | 11 | 492.13 | 19696.01 | | | | | | |